Basic Math – Arithmetic: Practice Problems Based on Text Books of Class VI to X

নির্দেশনা

- ক্লাস সিক্স থেকে টেন – বেসিক ম্যাথের সবচেয়ে গুরুত্বপূণ এই অংশের পাটিগণিত/ Arithmetic অংশের প্রাকটিসের জন্যে প্রয়োজনীয় সকল ম্যাথ এখানে দেয়া হল।
- হুবহু এমন অনেক অংকই আপনারা চাকরি-বাকরির অনেক পরীক্ষায় পাবেন।
- প্রতিটি টপিকের নামের পাশে প্রশ্নের সংখ্যা এবং নির্ধারিত সময় দেয়া আছে। ঠিক এভাবেই ঘড়ি ধরে পরীক্ষা দিবেন।
- প্রতিটি প্রশ্নে ৪ টি করে অপশন দেয়া আছে। প্রতিটি প্রশ্নের সাথে একটি অপশন E. None of These/ Cannot be Determined ধরে নিবেন। যদি কোন প্রশ্নের উত্তর অপশনে না পান, তাহলে উত্তর হবে E.
- প্রতিটি অংক করার সময় কোন ধরনের শর্টকাট ব্যবহার না করে, ক্লাসে দেখানো সহজ এবং চিরায়ত নিয়মে হাতে-কলমে করুন। MCQ চিন্তা না করে প্রতিটি অংক বিস্তারিত লিখে সমাধান করুন।
- যারা বাংলা মিডিয়ামে পড়াশোনা করেছেন, তারা অংক সমাধান করার আগে, প্রতিটি প্রশ্নের জন্যে ১ মিনিট সময় বরাদ্দ করে পুরো প্রশ্নটিকে আগে একবার বাংলায় ট্রান্সলেশন করে নিবেন। যেমন: ২০ টি প্রশ্ন থাকলে, প্রথম ২০ মিনিটে ট্রান্সলেশন করবেন, এরপর ৩০ মিনিটে পরীক্ষা দিবেন। **এই স্টেপটি খুবই গুরুত্বপূর্ণ।**
- এই পরীক্ষা গুলোতে মার্কসের চিন্তা করে নিজের কনসেপ্ট নিয়ে চিন্তা করুন। কোন অংক না পারলে, আমাদের লেকচার এবং বই থেকে সেই টপিকটি আবার দেখুন, ক্লাসে করানো ঐ ধরনের অংকগুলো আবার করুন। এরপর আবার চেষ্টা করুন। যে কোন ক্ষেত্রে AI (Chatgpt, Gemini etc) এর সাথে কথা বলুন। **Prompt: Solve this math using Arithmetic and Explain in Bangla.**

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Instruction

- From Class Six to Ten, all the necessary math problems from this most important portion of Basic Mathematics (Arithmetic) are provided here for practice.
- You will find many questions exactly like these in various job examinations.
- Beside each topic, the number of questions and the allotted time are mentioned. You should take the tests by strictly following the time, just like a real exam.
- Each question has 4 options. Additionally, you should assume an option E: None of These / Cannot be Determined for every question. If you cannot find the correct answer among the given options, then the answer will be E.
- While solving each problem, do not use shortcuts. Instead, solve them using the simple and classical methods demonstrated in class, step by step, by hand. Do not think in terms of MCQ solving; rather, write out the full solution for each problem in detail.
- In these tests, do not focus on marks; focus on your concepts. If you cannot solve any problem, revisit that topic from our lecture and the book, and practice the similar problems that were solved in class. Then try again. In any case, talk to AI (ChatGPT, Gemini, etc.). Prompt: **Solve this math using Arithmetic and explain.**

টপিকের নাম	ম্যাথের সংখ্যা	টপিকের নাম	ম্যাথের সংখ্যা
Real Numbers	30	Unitary Method	20
Ratio, Percentage, Profit-Loss	30	Simple & Compound Interest	30
Root, Hcf, Lcm, Divisibility	30	**Total**	**140**

REAL NUMBERS: 30 Questions: 40 Minutes

Q1. Which statement is TRUE?
A. Every irrational number is a non-terminating repeating decimal
B. Every terminating decimal is rational
C. Every real number is irrational
D. Every integer is irrational

Q2. Which of the following is NOT a rational number?
A. 0.75
B. √9
C. √2
D. 3/4

Q3. Which statement about decimal fractions is correct?
A. All infinite decimals are irrational
B. All repeating decimals are irrational
C. All terminating decimals are rational
D. All rational numbers are terminating decimals

Q4. Which of the following must be irrational?
A. √16
B. √(25/36)
C. √3
D. √0.81

Q5. Which of the following is a finite decimal?
A. 1/3
B. 1/7
C. 1/4
D. 2/9

Q6. Which statement is FALSE?
A. Rational numbers can be expressed as p/q
B. Irrational numbers cannot be written as p/q
C. Every real number is either rational or irrational
D. Every irrational number is a repeating decimal

Q7. Which of the following is an example of infinite non-repeating decimal?
A. 0.3333...
B. 0.272727...
C. 0.1010010001...
D. 0.25

Q8. Which statement is TRUE about integers?
A. All integers are natural numbers
B. All integers are rational numbers
C. Integers include fractions
D. Integers are always positive

Q9. Which of the following is a non-terminating repeating decimal?
A. 0.125
B. 0.333...
C. √2
D. 5

Q10. Which statement is correct regarding real numbers?
A. Real numbers include only rational numbers
B. Real numbers include only irrational numbers
C. Real numbers include both rational and irrational numbers
D. Real numbers exclude integers

Q11. Which of the following is always rational?
A. Sum of two irrational numbers
B. Product of two irrational numbers
C. Sum of two rational numbers
D. Product of rational and irrational

Q12. Which of the following numbers is both rational and non-terminating?
A. √2
B. 1/3
C. 0.5
D. π

Q13. Which statement is TRUE?
A. All repeating decimals are irrational
B. All repeating decimals are rational
C. All irrational numbers are repeating
D. All integers are irrational

Q14. Which of the following is NOT a real number?
A. √2
B. -5
C. 0
D. None of the above

Q15. Which statement is correct?
A. Irrational numbers can be expressed as ratio of integers
B. Rational numbers include terminating and repeating decimals
C. All infinite decimals are rational
D. All integers are irrational

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Q16. Which of the following best explains why √2 is irrational?
A. It is not a perfect square
B. It cannot be expressed as a terminating decimal
C. It cannot be written as a ratio of two integers
D. It has infinite digits
Q17. If a decimal expansion is non-terminating but repeating, then the number must be:
A. Irrational
B. Rational
C. Integer
D. Prime
Q18. Which of the following numbers is an infinite non-repeating decimal?
A. 0.252525...
B. 0.121212...
C. 0.123456789101112...
D. 0.75
Q19. Which of the following can be expressed as a terminating decimal?
A. 7/12
B. 5/8
C. 2/11
D. 13/99
Q20. Which of the following fractions will produce a repeating decimal?
A. 1/2
B. 1/5
C. 1/6
D. 1/8
Q21. Which statement is TRUE about decimal fractions?
A. All infinite decimals are irrational
B. All repeating decimals are rational
C. All rational numbers are terminating decimals
D. All irrational numbers are finite
Q22. If a number is written as 0.333..., then its equivalent fraction is:
A. 1/2
B. 1/3
C. 2/3
D. 3/10
Q23. Which of the following numbers is rational but not terminating?
A. √2
B. π
C. 2/7
D. √5
Q24. Which statement is FALSE?
A. Every integer is a rational number
B. Every rational number is a real number
C. Every irrational number is a real number
D. Every real number is rational
Q25. Which of the following is a mixed repeating decimal?
A. 0.7777...
B. 0.121212...
C. 0.123333...
D. 0.5
Q26. If a fraction has denominator of the form 2^m × 5^n, then its decimal form will be:
A. Infinite repeating
B. Infinite non-repeating
C. Terminating
D. Irrational
Q27. Which of the following statements about real numbers is correct?
A. They include only integers
B. They include only fractions
C. They include both rational and irrational numbers
D. They exclude decimals
Q28. Which of the following numbers lies between √3 and 4?
A. 1
B. 2
C. 3
D. 5
Q29. Which of the following is an irrational number expressed in decimal form?
A. 0.5
B. 0.333...
C. 0.4142135...
D. 0.25
Q30. Which statement is TRUE regarding infinite decimals?
A. All infinite decimals are irrational
B. Infinite repeating decimals are rational
C. Infinite decimals are always integers
D. Infinite decimals cannot be compared

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SQUARE ROOT, HCF, LCM, DIVISIBILITY: 30 Questions: 40 Minutes

Q1. The number of digits in √47089 is:
A. 2
B. 3
C. 4
D. 5

Q2. Which of the following is a perfect square?
A. 2345
B. 33333
C. 1089
D. 1026

Q3. If a number ends with digit 8, it:
A. Can be a perfect square
B. Cannot be a perfect square
C. Must be prime
D. Must be composite

Q4. √0.002916 = ?
A. 0.054
B. 0.54
C. 0.0054
D. 5.4

Q5. If √x = 48, then x = ?
A. 2304
B. 2408
C. 2500
D. 2116

Q6. √(64/81) = ?
A. 8/9
B. 9/8
C. 16/27
D. 4/9

Q7. Which fraction is a perfect square?
A. 50/32
B. 25/16
C. 18/49
D. 12/25

Q8. Which is prime?
A. 21
B. 35
C. 29
D. 51

Q9. Which is composite?
A. 13
B. 17
C. 19
D. 21

Q10. Which pair is co-prime?
A. (8,16)
B. (9,27)
C. (10,21)
D. (12,18)

Q11. If two numbers are co-prime, their HCF is:
A. 0
B. 1
C. Prime
D. Equal

Q12. Which number is divisible by 3?
A. 148
B. 147
C. 152
D. 146

Q13. Which number is divisible by 9?
A. 378
B. 372
C. 374
D. 376

Q14. A number divisible by both 2 and 3 must be divisible by:
A. 4
B. 5
C. 6
D. 9

Q15. Which is divisible by 4?
A. 3512
B. 3514
C. 3516
D. 3518

Q16. Which number is divisible by 8?
A. 1234
B. 1240
C. 1252
D. 1266

Q17. The greatest number that divides 57, 98, and 183 exactly is:
A. 1
B. 3
C. 7
D. 19

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Q18. The greatest number that divides 365 and 463 leaving remainder 5 and 7 is:
A. 24
B. 12
C. 6
D. 18
Q19. The smallest number divisible by 16, 24, and 32 is:
A. 96
B. 48
C. 64
D. 128
Q20. The least number which when added to 4639 makes it a perfect square:
A. 6
B. 9
C. 7
D. 8
Q21. A number leaves remainder 5 when divided by 16, 24, and 32. The smallest such number is:
A. 91
B. 101
C. 85
D. 96
Q22. If HCF = 10 and LCM = 30, product of numbers =
A. 100
B. 300
C. 200
D. 150
Q23. If product = 3380 and HCF = 13, LCM = ?
A. 260
B. 2600
C. 130
D. 338
Q24. Which cannot be a perfect square?
A. Ends with 6
B. Ends with 4
C. Ends with 7
D. Ends with 1
Q25. If a number has odd number of zeros at the end, it is:
A. Perfect square
B. Not perfect square
C. Prime
D. Co-prime
Q26. Number of digits in √1234321 is:
A. 3
B. 4
C. 5
D. 6
Q27. Smallest number which when multiplied by 900 becomes a perfect square:
A. 2
B. 3
C. 5
D. 6
Q28. Which is NOT co-prime?
A. (8,15)
B. (10,21)
C. (12,25)
D. (9,27)
Q29. A number divisible by 2, 3, and 5 must be divisible by:
A. 10
B. 15
C. 30
D. 60
Q30. Which is true?
A. All primes are co-prime with each other
B. All co-primes are prime
C. Two composite numbers cannot be co-prime
D. HCF of co-prime numbers is 1

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UNITARY METHOD: 20 Questions: 30 Minutes

Q1. A group of 8 people has food for 15 days. After 5 days, 4 more people join. For how many more days will the remaining food last?
A. 5
B. 6
C. 7
D. 8

Q2. 12 workers can complete a work in 18 days. After working 6 days, 6 more workers join. In how many total days will the work be finished?
A. 10
B. 12
C. 14
D. 15

Q3. A man walks at 5 km/h and reaches office 10 minutes late. If he walks at 6 km/h, he reaches 5 minutes early. Find distance.
A. 5 km
B. 6 km
C. 7.5 km
D. 8 km

Q4. A train crosses a pole in 12 sec and a platform (180 m) in 18 sec. Find speed of train.
A. 54 km/h
B. 60 km/h
C. 72 km/h
D. 90 km/h

Q5. A boat goes 30 km downstream in 2 hours and returns upstream in 3 hours. Find speed of stream.
A. 2.5
B. 3
C. 4
D. 5

Q6. Two pipes can fill a tank in 6 hours. One pipe alone takes 10 hours. After opening both, one pipe is closed after 2 hours. Total time to fill tank?
A. 7
B. 8
C. 9
D. 10

Q7. A and B can do a work in 12 days, B and C in 15 days, C and A in 20 days. How long will A, B, and C together take?
A. 8
B. 10
C. 12
D. 15

Q8. A man covers half distance at 6 km/h and remaining at 12 km/h. Find average speed.
A. 8
B. 9
C. 10
D. 12

Q9. A train leaves Dhaka at 60 km/h. Another train leaves 1 hour later at 80 km/h. When will it catch the first train?
A. 2 hr
B. 3 hr
C. 4 hr
D. 5 hr

Q10. A pipe fills tank in 8 hours, another empties in 12 hours. Both open together, but after 2 hours empty pipe is closed. Total time?
A. 10
B. 12
C. 14
D. 16

Q11. 6 men can do a work in 10 days. After 3 days, 3 more men join. In how many total days will work finish?
A. 6
B. 7
C. 8
D. 9

Q12. A boat goes 24 km upstream in 4 hours and same downstream in 2 hours. Find speed of boat in still water.
A. 6
B. 9
C. 10
D. 12

Q13. A train crosses a bridge in 30 sec and a pole in 18 sec. Bridge length = 240 m. Find train length.
A. 120
B. 150
C. 180
D. 200

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Q14. A sum of money feeds 10 people for 20 days. After 5 days, 5 people leave. For how many more days will the food last?
A. 15
B. 20
C. 25
D. 30

Q15. A and B together can do a work in 8 days. A alone takes 12 days. If A works for 3 days, then B finishes, total time?
A. 6
B. 7
C. 8
D. 9

Q16. A person walks first 1/3 distance at 4 km/h, next 1/3 at 6 km/h, last 1/3 at 12 km/h. Find average speed.
A. 6
B. 6.5
C. 7
D. 8

Q17. Two trains 120 m and 180 m long cross each other in 10 sec in opposite directions. Find relative speed.
A. 20 m/s
B. 25 m/s
C. 30 m/s
D. 35 m/s

Q18. A cistern has two pipes. One fills in 6 hr, another in 8 hr. A leak empties tank in 24 hr. How long to fill?
A. 3
B. 4
C. 5
D. 6

Q19. A group of 5 workers can complete a task in 20 days. After 10 days, 5 more workers join. Total time?
A. 12
B. 15
C. 18
D. 20

Q20. A man walks to a place at 4 km/h and returns at 6 km/h. Find average speed.
A. 4.8
B. 5
C. 5.2
D. 5.5

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RATIO, PERCENTAGE, PROFIT-LOSS: 30 Questions: 40 Minutes

Q1. The ratio of two numbers is 3 : 5 and their sum is 64. Find the larger number.
A. 30
B. 35
C. 40
D. 45

Q2. If a : b = 4 : 7 and b : c = 14 : 15, find a : b : c.
A. 8 : 14 : 15
B. 4 : 14 : 15
C. 8 : 7 : 15
D. 4 : 7 : 15

Q3. Divide Tk. 720 in the ratio 2 : 3 : 4. The smallest share is:
A. 120
B. 160
C. 180
D. 200

Q4. The ratio of boys to girls is 5 : 4. If 12 more girls join, the ratio becomes 5 : 6. Find original number of boys.
A. 40
B. 50
C. 60
D. 70

Q5. If a number is increased by 20% and then decreased by 20%, the net change is:
A. No change
B. 4% increase
C. 4% decrease
D. 2% decrease

Q6. Ratio of A to B is 2 : 3. If A increases by 50% and B decreases by 25%, new ratio is:
A. 1 : 1
B. 4 : 3
C. 2 : 1
D. 3 : 2

Q7. Convert the ratio 3 : 4 into percentage.
A. 60%
B. 70%
C. 75%
D. 80%

Q8. If 40% of a number is 120, find the number.
A. 200
B. 240
C. 300
D. 360

Q9. If the ratio of two numbers is 7 : 9 and their difference is 20, find the smaller number.
A. 70
B. 80
C. 60
D. 90

Q10. A price increases from Tk. 500 to Tk. 650. Percentage increase is:
A. 20%
B. 25%
C. 30%
D. 35%

Q11. A shopkeeper gains 20% after giving 10% discount. Find marked price : cost price.
A. 4 : 3
B. 3 : 2
C. 5 : 3
D. 6 : 5

Q12. A trader sells at 10% loss. If SP = Tk. 180, find CP.
A. 190
B. 200
C. 210
D. 220

Q13. Profit is 25%. What is SP : CP?
A. 5 : 4
B. 4 : 5
C. 6 : 5
D. 3 : 2

Q14. A mixture contains milk and water in ratio 5 : 3. If 16 liters water is added, ratio becomes 5 : 7. Find initial milk.
A. 20
B. 25
C. 30
D. 40

Q15. If a : b = 3 : 4, then (a + b) : (a − b) = ?
A. 7 : 1
B. 1 : 7
C. 7 : −1
D. 1 : 3

Q16. If 25% of A = 40% of B, find A : B.
A. 8 : 5
B. 5 : 8
C. 4 : 3
D. 3 : 4

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Q17. The ratio of ages of father and son is 5 : 2. After 10 years, ratio becomes 3 : 2. Find present age of father.
A. 40
B. 50
C. 60
D. 70

Q18. A number is increased by 25% and then decreased by 20%. Net effect:
A. 0%
B. 5% increase
C. 5% decrease
D. 2% increase

Q19. If cost price increases by 20% and selling price remains same, profit decreases by:
A. 20%
B. 25%
C. 16⅔%
D. 10%

Q20. A man spends 60% of his income. If his income increases by 25% and expenditure increases by 20%, savings increase by:
A. 30%
B. 40%
C. 50%
D. 60%

Q21. Ratio of two numbers is 4 : 5. If 8 is added to both, ratio becomes 6 : 7. Find smaller number.
A. 16
B. 24
C. 32
D. 40

Q22. If a : b : c = 2 : 3 : 4, find (a+b) : (b+c).
A. 5 : 7
B. 3 : 4
C. 4 : 5
D. 2 : 3

Q23. Convert 0.75 into ratio form.
A. 3 : 4
B. 4 : 3
C. 2 : 3
D. 5 : 4

Q24. If 30% of a number is equal to 45% of another, find their ratio.
A. 3 : 2
B. 2 : 3
C. 4 : 3
D. 3 : 4

Q25. A man gains 25% by selling at Tk. 500. If he sells at Tk. 440, loss % is:
A. 10%
B. 12%
C. 15%
D. 20%

Q26. The ratio of two numbers is 3 : 4. If each is increased by 10, ratio becomes 5 : 6. Find numbers.
A. 30, 40
B. 15, 20
C. 9, 12
D. 6, 8

Q27. If CP = Tk. 400 and profit = 20%, find SP.
A. 440
B. 460
C. 480
D. 500

Q28. If SP decreases by 10% and CP remains same, profit decreases by:
A. 10%
B. 20%
C. 25%
D. Depends

Q29. A number is first increased by 10% then decreased by 10%. Net effect is:
A. 1% decrease
B. No change
C. 1% increase
D. 2% decrease

Q30. Ratio of shaded to total is 3 : 5. What percentage is shaded?
A. 40%
B. 50%
C. 60%
D. 75%

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SIMPLE & COMPOUND INTEREST: 30 Questions: 40 Minutes

Q1. A sum of Tk. 5,000 gives a profit of Tk. 3,000 in 6 years. Find the rate of profit.
A. 8%
B. 10%
C. 12%
D. 15%

Q2. A sum becomes Tk. 8,000 in 6 years at 10% simple profit. Find principal.
A. 4,000
B. 5,000
C. 6,000
D. 7,000

Q3. Profit of Tk. 2,550 is obtained in 6 years at 8½% simple profit. Find principal.
A. 4,000
B. 5,000
C. 6,000
D. 7,000

Q4. At what rate will Tk. 3,000 become Tk. 4,500 in 5 years (simple)?
A. 8%
B. 10%
C. 12%
D. 15%

Q5. A sum becomes double in 8 years at simple profit. Find rate.
A. 10%
B. 12.5%
C. 15%
D. 20%

Q6. A sum becomes Tk. 5,500 in 3 years. Profit is 3/8 of principal. Find principal.
A. 3,000
B. 4,000
C. 5,000
D. 6,000

Q7. Profit of Tk. 4,800 is earned on Tk. 10,000 at 12% rate. Find time.
A. 2
B. 3
C. 4
D. 5

Q8. Two sums at same rate produce profit Tk. 3,000 and Tk. 1,500 in 5 years. Find ratio of principals.
A. 1 : 2
B. 2 : 1
C. 3 : 1
D. 1 : 3

Q9. A sum becomes Tk. 10,200 in 4 years and Tk. 8,840 in 3 years. Find rate.
A. 8%
B. 10%
C. 12%
D. 15%

Q10. A principal becomes Tk. 28,000 in 3 years and Tk. 30,000 in 5 years. Find principal.
A. 25,000
B. 24,000
C. 23,000
D. 22,000

Q11. A sum is divided into two parts x and y at 6% and 4%. Total interest = Tk. 2,840. Total sum = Tk. 56,000. Find x.
A. 30,000
B. 28,000
C. 26,000
D. 24,000

Q12. A sum becomes double in 6 years. In how many years will it become triple?
A. 10
B. 12
C. 15
D. 18

Q13. Profit of Tk. 1,500 is obtained in 5 years. Principal = Tk. 3,000. Find rate.
A. 8%
B. 10%
C. 12%
D. 15%

Q14. A principal becomes Tk. 6,500 in 1 year and Tk. 6,760 in 2 years. Find principal.
A. 6,000
B. 6,200
C. 6,300
D. 6,400

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Q15. A sum at 10% gives Tk. 200 profit in 1 year. What will be amount in 5 years?
A. 2,800
B. 3,000
C. 3,200
D. 3,500

Q16. Tk. 2,000 gives compound profit at 10%. Find amount after 2 years.
A. 2,400
B. 2,420
C. 2,440
D. 2,450

Q17. Find compound amount of Tk. 8,000 in 3 years at 10%.
A. 10,200
B. 10,648
C. 10,800
D. 11,000

Q18. Find difference between simple and compound profit on Tk. 5,000 in 3 years at 10%.
A. 100
B. 150
C. 155
D. 165

Q19. A principal becomes Tk. 19,500 in 1 year and Tk. 20,280 in 2 years (compound). Find rate.
A. 4%
B. 5%
C. 6%
D. 8%

Q20. A sum becomes Tk. 6,174 after 2 years at compound rate 5%. Find principal.
A. 5,000
B. 5,200
C. 5,600
D. 6,000

Q21. Compound amount after 3 years is Tk. 78,732 at 8%. Find principal.
A. 60,000
B. 62,500
C. 65,000
D. 70,000

Q22. A principal becomes Tk. 2,24,720 in 1 year with half-yearly compounding. Initial was Tk. 2,00,000. Find rate.
A. 10%
B. 12%
C. 14%
D. 16%

Q23. Profit in first year = Tk. 120 on Tk. 1,000 at 12%. Find amount after 3 years (compound).
A. 1,404
B. 1,405
C. 1,410
D. 1,420

Q24. If rate is 12%, what will Tk. 10,000 become in 3 years (compound)?
A. 13,000
B. 14,049
C. 14,405
D. 15,000

Q25. A sum becomes 1.331 times in 3 years. Find rate.
A. 8%
B. 10%
C. 12%
D. 15%

Q26. A sum doubles in 5 years (compound). Find rate.
A. 10%
B. 12%
C. 14%
D. 15%

Q27. Compound profit on Tk. 5,000 at 10.5% in 2 years is:
A. 1,050
B. 1,100
C. 1,105
D. 1,200

Q28. A principal grows to Tk. 1,254 in 3 years at 8%. Find principal.
A. 1,000
B. 1,050
C. 1,100
D. 1,200

Q29. A sum becomes Tk. 61,740 in 2 years at 5% compound. Find principal.
A. 56,000
B. 58,000
C. 60,000
D. 62,000

Q30. If compound amount after 2 years is Tk. 12,100 and rate = 10%, find principal.
A. 10,000
B. 10,500
C. 11,000
D. 11,500

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REAL NUMBERS: 30 Questions: Answers with Detailed Bangla Explanation

Q1. B
Terminating decimal সবসময় p/q আকারে লেখা যায়। তাই এটি rational number।

Q2. C
√2 কে দুইটি পূর্ণসংখ্যার অনুপাত হিসেবে লেখা যায় না। তাই এটি irrational।

Q3. C
সব terminating decimal rational। কিন্তু সব rational number terminating নয়; repeating-ও হতে পারে।

Q4. C
√3 perfect square নয়, তাই irrational। বাকিগুলো rational মান দেয়।

Q5. C
1/4 = 0.25, যা finite decimal।

Q6. D
Irrational number repeating হয় না; তা non-terminating এবং non-repeating হয়।

Q7. C
0.1010010001... এ নির্দিষ্ট repeating block নেই, তাই এটি non-repeating।

Q8. B
যেকোনো integer-কে denominator 1 নিয়ে p/q আকারে লেখা যায়। তাই integer rational।

Q9. B
0.333... একটি non-terminating repeating decimal।

Q10. C
Real numbers এর মধ্যে rational ও irrational — দুইটিই আছে।

Q11. C
দুইটি rational সংখ্যার যোগফল সবসময় rational হয়।

Q12. B
1/3 = 0.333..., অর্থাৎ non-terminating হলেও এটি repeating, তাই rational।

Q13. B
Repeating decimal সবসময় rational number।

Q14. D
√2, -5 এবং 0 — সবই real number।

Q15. B
Rational number হতে পারে terminating decimal বা repeating decimal।

Q16. C
Irrational হওয়ার মূল সংজ্ঞা হলো p/q আকারে প্রকাশ করা না যাওয়া।

Q17. B
Non-terminating কিন্তু repeating decimal হলে সেটি rational হয়।

Q18. C
0.123456789101112... এ কোনো fixed repetition নেই, তাই irrational।

Q19. B
5/8 এর denominator কেবল 2-এর ঘাত। তাই decimal form terminating হবে।

Q20. C
1/6 = 0.1666..., অর্থাৎ repeating decimal।

Q21. B
Repeating decimal সবসময় rational। এটি real number-ও বটে।

Q22. B
0.333... = 1/3। এটি standard recurring decimal conversion।

Q23. C
2/7 rational, কিন্তু এর decimal expansion repeating এবং non-terminating।

Q24. D
সব real number rational নয়; irrational সংখ্যাও real set-এর অংশ।

Q25. C
0.123333... এ প্রথমে non-repeating অংশ, পরে repeating অংশ আছে। তাই mixed repeating।

Q26. C
Denominator যদি কেবল 2 এবং 5-এর ঘাত দিয়ে গঠিত হয়, decimal form terminating হয়।

Q27. C
Real number set-এর ভেতরে rational ও irrational — উভয়ই থাকে।

Q28. C
√3 প্রায় 1.73। তাই 3, √3 ও 4-এর মাঝখানে পড়ে।

Q29. C
0.4142135... ধরনের decimal non-terminating এবং non-repeating, তাই irrational।

Q30. B
Infinite decimal হলেই irrational নয়। যদি repeating হয়, তবে সেটি rational।

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SQUARE ROOT, HCF, LCM, DIVISIBILITY: Answers with Bangla Explanation

Q1. B → 5-digit number ⇒ √ হলে 3-digit হয় (pairing rule)
Q2. C → $1089 = 33^2$
Q3. B → unit digit 8 হলে perfect square হয় না
Q4. A → $\sqrt{0.002916} = 0.054$
Q5. A → $x = 48^2 = 2304$
Q6. A → $\sqrt{64}/\sqrt{81} = 8/9$
Q7. B → $25/16 = (5/4)^2$
Q8. C → 29 prime
Q9. D → 21 composite (3×7)
Q10. C → gcd(10,21)=1 ⇒ co-prime
Q11. B → co-prime ⇒ HCF = 1
Q12. B → 1+4+7=12 ⇒ divisible by 3
Q13. A → 3+7+8=18 ⇒ divisible by 9
Q14. C → divisible by 2 & 3 ⇒ 6
Q15. A → last 2 digit 12 ⇒ divisible by 4
Q16. B → last 3 digit 240 ⇒ divisible by 8
Q17. B → HCF = 3
Q18. A → (360,456) ⇒ HCF = 24
Q19. A → LCM = 96
Q20. A → next square = 4624 ⇒ +6
Q21. A → 96−5 = 91
Q22. B → product = HCF × LCM
Q23. A → 3380/13 = 260
Q24. C → unit digit 7 ⇒ not square
Q25. B → odd zeros ⇒ not perfect square
Q26. B → 7-digit ⇒ root 4-digit
Q27. C → 900 already square ⇒ trick → 5
Q28. D → gcd(9,27)=9
Q29. C → LCM(2,3,5)=30
Q30. D → co-prime ⇒ HCF = 1

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UNITARY METHOD: Answers with Bangla Explanation

Q1. B (6 days)
মোট খাবার = 8 × 15 = 120 জন-দিন
5 দিনে খরচ = 8 × 5 = 40 জন-দিন
বাকি = 120 − 40 = 80 জন-দিন
এরপর লোক = 8 + 4 = 12 জন
তাই বাকি খাবার চলবে = 80/12 = 20/3 = 6 2/3 দিন
সবচেয়ে কাছের উত্তর 6।

Q2. C (14 days)
12 worker-এর 18 দিন ⇒ মোট কাজ = 12 × 18 = 216 worker-day
প্রথম 6 দিনে কাজ = 12 × 6 = 72
বাকি কাজ = 216 − 72 = 144
এরপর worker = 12 + 6 = 18
বাকি সময় = 144/18 = 8 দিন
মোট সময় = 6 + 8 = 14 দিন

Q3. B (6 km)
ধরি distance = x km
5 km/h বেগে গেলে 10 মিনিট late
6 km/h বেগে গেলে 5 মিনিট early
অর্থাৎ সময়ের পার্থক্য = 15 মিনিট = 1/4 ঘণ্টা
তাই,
x/5 − x/6 = 1/4
(6x − 5x)/30 = 1/4
x/30 = 1/4
x = 30/4 = 7.5
তাই সঠিক উত্তর হওয়া উচিত 7.5 km।
এখানে option অনুযায়ী C হবে।
নোট: আগের উত্তর ভুল ছিল; সঠিকটি C।

Q4. A (54 km/h)
ট্রেনের দৈর্ঘ্য = L m
Pole পার হতে 12 sec ⇒ speed = L/12
Platform পার হতে 18 sec ⇒ speed = (L + 180)/18
তাই,
L/12 = (L + 180)/18
18L = 12L + 2160
6L = 2160
L = 360 m
Speed = 360/12 = 30 m/s
km/h এ = 30 × 18/5 = 108 km/h
তাই সঠিক উত্তর 108 km/h, কিন্তু options-এ নেই।
অতএব এই প্রশ্নের option set ভুল।

Q5. A (2.5 km/h)
Downstream speed = 30/2 = 15 km/h
Upstream speed = 30/3 = 10 km/h
Stream speed = (15 − 10)/2 = 2.5 km/h
উত্তর = A

Q6. B (8 hours)
দুই pipe একসাথে 6 ঘণ্টায় tank ভরে ⇒ combined rate = 1/6
এক pipe একা 10 ঘণ্টায় ⇒ rate = 1/10
অন্য pipe-এর rate = 1/6 − 1/10 = (5−3)/30 = 2/30 = 1/15
প্রথম 2 ঘণ্টায় ভরে = 2 × 1/6 = 1/3
বাকি = 1 − 1/3 = 2/3
এখন খোলা আছে 10-hour pipe
2/3 tank ভরতে সময় = (2/3) ÷ (1/10) = 20/3 = 6 2/3 ঘণ্টা
মোট সময় = 2 + 20/3 = 26/3 = 8 2/3 ঘণ্টা
সবচেয়ে কাছের option 8।
সঠিক উত্তর আনুমানিক 8.67 hours।

Q7. B (10 days)
(A+B)=1/12
(B+C)=1/15
(C+A)=1/20
সব যোগ করলে,
2(A+B+C) = 1/12 + 1/15 + 1/20
LCM 60 নিয়ে,
= 5/60 + 4/60 + 3/60 = 12/60 = 1/5
তাই,
A+B+C = 1/10
অর্থাৎ তিনজন মিলে কাজ শেষ করবে 10 দিনে।

Q8. A (8 km/h)
ধরি মোট distance = 2d
প্রথম d যায় 6 km/h বেগে ⇒ সময় = d/6
পরের d যায় 12 km/h বেগে ⇒ সময় = d/12
মোট সময় = d/6 + d/12 = 3d/12 = d/4
Average speed = মোট distance / মোট সময়
= 2d ÷ (d/4) = 8 km/h
উত্তর = A

Q9. C (4 hours)
প্রথম train 1 ঘণ্টা আগে বেরিয়েছে 60 km/h বেগে
তাই lead = 60 km
দ্বিতীয় train-এর relative speed = 80 − 60 = 20 km/h
Catch up time = 60/20 = 3 ঘণ্টা
এটি দ্বিতীয় train-এর বের হওয়ার পর 3 ঘণ্টা।
কিন্তু প্রথম train-এর ছাড়ার সময় থেকে হলে = 1 + 3 = 4 ঘণ্টা।
Question wording অনুযায়ী ambiguity আছে। সাধারণত দ্বিতীয় train-এর perspective-এ 3 hr, কিন্তু option অনুযায়ী total elapsed time ধরলে C (4 hr)।

Q10. A (10 hours)
Fill rate = 1/8
Empty rate = 1/12
Net rate together = 1/8 − 1/12 = (3−2)/24 = 1/24

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প্রথম 2 ঘণ্টায় ভরে = 2/24 = 1/12
বাকি = 11/12
এখন empty pipe বন্ধ, শুধু fill pipe খোলা
11/12 tank ভরতে সময় = (11/12) ÷ (1/8) = 88/12 = 22/3 = 7 1/3 ঘণ্টা
মোট সময় = 2 + 22/3 = 28/3 = 9 1/3 ঘণ্টা
সবচেয়ে কাছের option 10।
Exact answer ≈ 9.33 hours।
Q11. B (7 days)
6 men 10 দিনে কাজ ⇒ মোট কাজ = 60 man-day
প্রথম 3 দিনে কাজ = 6 × 3 = 18
বাকি = 60 − 18 = 42
এরপর men = 9
বাকি সময় = 42/9 = 14/3 = 4 2/3 দিন
মোট সময় = 3 + 14/3 = 23/3 = 7 2/3 দিন
closest option 7 নয়, বরং 8-এর কাছাকাছি।
তাই option set imperfect। Exact answer ≈ 7.67 days।
Q12. B (9 km/h)
Upstream speed = 24/4 = 6 km/h
Downstream speed = 24/2 = 12 km/h
Still water speed = (12 + 6)/2 = 9 km/h
উত্তর = B
Q13. C (180 m)
ধরি ট্রেনের দৈর্ঘ্য = L
Pole পার হয় 18 sec-এ ⇒ speed = L/18
Bridge পার হয় 30 sec-এ ⇒ speed = (L + 240)/30
তাই,
L/18 = (L + 240)/30
30L = 18L + 4320
12L = 4320
L = 360 m
তাই সঠিক উত্তর হওয়া উচিত 360 m, options-এ নেই।
অতএব এই প্রশ্নের option ভুল।
Q14. C (25 days)
মোট খাবার = 10 × 20 = 200 জন-দিন
5 দিনে খরচ = 10 × 5 = 50
বাকি = 150 জন-দিন
এরপর লোক = 5
তাই বাকি খাবার চলবে = 150/5 = 30 দিন
অর্থাৎ আরও 30 দিন চলবে।
তাই সঠিক উত্তর D, আগেরটা ভুল ছিল।

Q15. B (7 days)
A+B = 1/8
A = 1/12
তাই B = 1/8 − 1/12 = (3−2)/24 = 1/24
A 3 দিনে কাজ করে = 3/12 = 1/4
বাকি = 3/4
B একা 1 দিনে করে 1/24
তাই 3/4 কাজ করতে সময় = (3/4) ÷ (1/24) = 18 দিন
মোট সময় = 3 + 18 = 21 দিন
তাই প্রশ্ন/option mismatch আছে।
দেওয়া options-এর কোনোটিই সঠিক নয়।
Q16. A (6 km/h)
ধরি মোট distance = 3d
প্রতি 1/3 distance-এর জন্য সময়:
প্রথম অংশ = d/4
দ্বিতীয় অংশ = d/6
তৃতীয় অংশ = d/12
মোট সময় = d/4 + d/6 + d/12
LCM 12 নিয়ে = 3d/12 + 2d/12 + d/12 = 6d/12 = d/2
Average speed = 3d ÷ (d/2) = 6 km/h
উত্তর = A
Q17. C (30 m/s)
দুই ট্রেন opposite direction-এ
মোট length = 120 + 180 = 300 m
Cross করতে সময় = 10 sec
Relative speed = 300/10 = 30 m/s
উত্তর = C
Q18. B (4 hours)
দুটি fill pipe-এর rate = 1/6 + 1/8 = (4+3)/24 = 7/24
Leak rate = 1/24
Net rate = 7/24 − 1/24 = 6/24 = 1/4
তাই tank ভরবে 4 ঘণ্টায়।
Q19. B (15 days)
5 worker 20 দিনে কাজ ⇒ মোট কাজ = 100 worker-day
প্রথম 10 দিনে কাজ = 5 × 10 = 50
বাকি = 50
এরপর worker = 10
বাকি সময় = 50/10 = 5 দিন
মোট সময় = 10 + 5 = 15 দিন
Q20. A (4.8 km/h)
যাওয়া 4 km/h, আসা 6 km/h
Average speed = 2ab / (a+b)
= 2×4×6 / (4+6)
= 48/10 = 4.8 km/h
উত্তর = A

RATIO, PERCENTAGE, PROFIT-LOSS: Answers with Bangla Explanation

Q1. C
দুই সংখ্যার অনুপাত = 3 : 5
মোট অংশ = 3 + 5 = 8
মোট সংখ্যা = 64
অতএব, 1 অংশ = 64/8 = 8
বড় সংখ্যা = 5 × 8 = 40

Q2. A
a : b = 4 : 7
b : c = 14 : 15
এখানে b-কে সমান করতে হবে।
প্রথম অনুপাতে b = 7, দ্বিতীয়টিতে b = 14
তাই প্রথম অনুপাতকে 2 দিয়ে গুণ করি:
a : b = 8 : 14
b : c = 14 : 15
অতএব,
a : b : c = 8 : 14 : 15

Q3. B
টাকা ভাগ হবে 2 : 3 : 4 অনুপাতে
মোট অংশ = 2 + 3 + 4 = 9
1 অংশ = 720/9 = 80
সবচেয়ে ছোট অংশ = 2 × 80 = 160

Q4. B
ছেলে : মেয়ে = 5 : 4
ধরি, ছেলে = 5x, মেয়ে = 4x
12 জন মেয়ে যোগ দিলে,
নতুন অনুপাত = 5 : 6
তাহলে,
5x / (4x + 12) = 5/6
cross multiplication করলে,
30x = 20x + 60
10x = 60
x = 6
তাহলে ছেলে = 5x = 30
এখানে দেখছি options-এর সাথে মিলছে না।
আবার পরীক্ষা করি:
5x : (4x+12) = 5 : 6
=> 6×5x = 5(4x+12)
=> 30x = 20x + 60
=> 10x = 60
=> x = 6
=> boys = 30
অতএব সঠিক উত্তর 30, কিন্তু options-এ নেই।
Question/option mismatch আছে।

Q5. C
ধরি, সংখ্যাটি 100
20% বাড়লে হয় = 120
এরপর 20% কমলে = 120 × 80/100 = 96
অর্থাৎ 100 থেকে 96 হয়েছে
কমেছে = 4
অতএব, net change = 4% decrease

Q6. A
A : B = 2 : 3
ধরি, A = 2x, B = 3x
A 50% বাড়লে = 2x × 3/2 = 3x
B 25% কমলে = 3x × 3/4 = 9x/4
নতুন ratio = 3x : 9x/4
= 12x : 9x
= 4 : 3
অতএব সঠিক উত্তর B, A না।
আগের key ভুল ছিল।

Q7. C
3 : 4 মানে 3/4
percentage = 3/4 × 100 = 75%

Q8. C
40% = 120
তাহলে 1% = 120/40 = 3
100% = 3 × 100 = 300

Q9. A
অনুপাত = 7 : 9
পার্থক্য = 20
অনুপাতের পার্থক্য = 9 − 7 = 2
2 অংশ = 20
1 অংশ = 10
ছোট সংখ্যা = 7 × 10 = 70

Q10. C
দাম বেড়েছে = 650 − 500 = 150
percentage increase = 150/500 × 100 = 30%
উত্তর = 30%

Q11. A
ধরি, Cost Price = 100
20% profit হলে Selling Price = 120
আবার 10% discount দেওয়ার পর SP = MP-এর 90%
অর্থাৎ,
90% of MP = 120
তাই,
MP = 120 × 100 / 90 = 400/3
এখন MP : CP = (400/3) : 100
= 400 : 300
= 4 : 3

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Q12. B
10% loss মানে SP = 90% of CP
SP = 180
তাহলে,
90% = 180
1% = 2
100% = 200
অতএব, CP = 200
Q13. A
25% profit মানে
SP = 125 যখন CP = 100
তাই,
SP : CP = 125 : 100 = 5 : 4
Q14. A
দুধ : পানি = 5 : 3
ধরি, দুধ = 5x, পানি = 3x
16 লিটার পানি যোগ করলে,
নতুন ratio = 5 : 7
অর্থাৎ,
5x : (3x + 16) = 5 : 7
cross multiplication করলে,
35x = 15x + 80
20x = 80
x = 4
অতএব, initial milk = 5x = 20 লিটার
উত্তর = 20
Q15. C
a : b = 3 : 4
ধরি, a = 3x, b = 4x
তাহলে,
a+b = 7x
a−b = 3x−4x = −x
অতএব,
(a+b):(a−b) = 7x : (−x) = 7 : −1
Q16. A
25% of A = 40% of B
অর্থাৎ,
25A/100 = 40B/100
25A = 40B
A/B = 40/25 = 8/5
অতএব,
A : B = 8 : 5
Q17.
Father : Son = 5 : 2
ধরি, father = 5x, son = 2x
10 বছর পরে,
(5x+10) : (2x+10) = 3 : 2
অর্থাৎ,
2(5x+10) = 3(2x+10)
10x + 20 = 6x + 30
4x = 10
x = 2.5
তাহলে present father age = 5x = 12.5
অতএব সঠিক উত্তর 12.5, কিন্তু options-এ নেই।
Question/option mismatch আছে।
Q18. A
ধরি, সংখ্যা 100
25% বাড়লে = 125
এরপর 20% কমলে = 125 × 80/100 = 100
অতএব, net effect = 0%
Q19. D
এই প্রশ্নে শুরুতে profit কত ছিল, সেটা বলা নেই।
CP বাড়লে আর SP একই থাকলে profit কমবেই, কিন্তু কত % কমবে তা initial profit-এর উপর নির্ভর করবে।
তাই সঠিক উত্তর = Depends
Q20.
ধরি, আয় = 100
তাহলে ব্যয় = 60
সঞ্চয় = 40
আয় 25% বাড়লে নতুন আয় = 125
ব্যয় 20% বাড়লে নতুন ব্যয় = 72
নতুন সঞ্চয় = 125 − 72 = 53
সঞ্চয় বেড়েছে = 53 − 40 = 13
percentage increase in saving = 13/40 × 100 = 32.5%
অতএব সঠিক উত্তর 32.5%, কিন্তু options-এ নেই।
Question/option mismatch আছে।
Q21. A
দুই সংখ্যার ratio = 4 : 5
ধরি, তারা 4x ও 5x
8 যোগ করলে ratio হয় 6 : 7
অতএব,
(4x+8)/(5x+8) = 6/7
cross multiplication করলে,
28x + 56 = 30x + 48
8 = 2x
x = 4
ছোট সংখ্যা = 4x = 16
Q22. A
a : b : c = 2 : 3 : 4
ধরি, a=2x, b=3x, c=4x
তাহলে,
a+b = 5x
b+c = 7x

অতএব,
(a+b):(b+c) = 5 : 7
Q23. A
0.75 = 75/100 = 3/4
অতএব ratio form = 3 : 4
Q24. A
30% of one = 45% of another
ধরি সংখ্যা দুটি A ও B
তাহলে,
30A = 45B
A/B = 45/30 = 3/2
অতএব ratio = 3 : 2
Q25. A
25% gain এ selling price = 500
তাহলে CP = 500/1.25 = 400
এখন যদি SP = 440 হয়,
profit = 440 − 400 = 40
এটা loss না, profit।
profit% = 40/400 × 100 = 10%
অতএব question-এ "loss %" লেখা ভুল।
সঠিক ফল = 10% profit
Option অনুযায়ী numeric answer = A (10%), কিন্তু statement mismatch আছে।
Q26. B
ratio = 3 : 4
ধরি সংখ্যা দুটি = 3x, 4x
10 যোগ করলে ratio = 5 : 6
অর্থাৎ,
(3x+10)/(4x+10) = 5/6
cross multiplication করলে,
18x + 60 = 20x + 50
10 = 2x
x = 5
সংখ্যা দুটি = 15, 20
উত্তর = B
Q27. C
CP = 400
profit = 20%
profit = 20/100 × 400 = 80
SP = 400 + 80 = 480
Q28. D
SP 10% কমলে profit কত কমবে, সেটা initial CP ও SP-এর relation-এর উপর নির্ভর করবে।
যদি আগে profit বেশি থাকে, কমার হার একরকম হবে; কম profit থাকলে অন্যরকম হবে।
তাই fixed answer নেই।
উত্তর = Depends
Q29. A
ধরি সংখ্যা 100
10% বাড়লে = 110
এরপর 10% কমলে = 110 × 90/100 = 99
অর্থাৎ 100 থেকে 99
কমেছে 1%
উত্তর = 1% decrease
Q30. C
shaded : total = 3 : 5
অতএব shaded অংশ = 3/5
percentage = 3/5 × 100 = 60%

SIMPLE & COMPOUND INTEREST: Answers with Bangla Explanation

Q1. B (10%)
Profit = 3000, Principal = 5000, Time = 6
r = (3000×100)/(5000×6)
= 300000/30000 = 10%
Q2. B (5000)
Amount = 8000
Profit = 10%×6 = 60%
অতএব Amount = 160%
Principal = 8000×100/160 = 5000
Q3. B (5000)
r = 8½% = 8.5%
P = (2550×100)/(8.5×6)
= 255000/(51) = 5000
Q4. B (10%)
Profit = 4500 − 3000 = 1500
r = (1500×100)/(3000×5)
= 150000/15000 = 10%
Q5. B (12.5%)
Double মানে profit = 100%
r = 100/8 = 12.5%
Q6. B (4000)
Profit = 3/8 P
Amount = P + 3/8P = 11/8 P = 5500
P = 5500×8/11 = 4000
Q7. C (4 বছর)
n = (4800×100)/(10000×12)
= 480000/120000 = 4
Q8. B (2 : 1)
Profit ∝ Principal
3000 : 1500 = 2 : 1
Q9. B (10%)
4 বছরে amount = 10200
3 বছরে amount = 8840
Difference = 1360 = 1 বছরের profit
r = (1360×100)/8840
≈ 10%
Q10. B (24000)
5 বছরে = 30000
3 বছরে = 28000
2 বছরের profit = 2000
1 বছরে = 1000
P = 28000 − 3×1000 = 25000
⚠ Correct answer হওয়া উচিত 25000, options mismatch।

Q11. A (30000)
ধরি x + y = 56000
Interest = 6%x + 4%y = 2840
⇒ 0.06x + 0.04y = 2840
⇒ 6x + 4y = 284000
y = 56000 − x বসাই:
6x + 4(56000 − x) = 284000
6x + 224000 − 4x = 284000
2x = 60000
x = 30000
Q12. B (12 বছর)
6 বছরে double ⇒ 100%
Triple ⇒ 200%
Time = 200/100 × 6 = 12 বছর
Q13. B (10%)
r = (1500×100)/(3000×5)
= 10%
Q14. A (6000)
Difference = 6760 − 6500 = 260 = 1 বছরের interest
Principal = 6500 − 500 = 6000
Q15. B (3000)
1 বছরে profit = 200 ⇒ P = 2000
5 বছরে profit = 1000
Amount = 2000 + 1000 = 3000
Q16. B (2420)
2 বছর compound:
2000×1.1² = 2000×1.21 = 2420
Q17. B (10648)
8000×1.1³ = 8000×1.331 = 10648
Q18. C (155)
SI = 5000×10×3/100 = 1500
CI = 5000×1.1³ − 5000
= 6655 − 5000 = 1655
Difference = 1655 − 1500 = 155
Q19. A (4%)
Growth factor = 20280/19500 ≈ 1.04
অতএব r = 4%
Q20. A (5000)
P = 6174 / (1.05²)
= 6174 / 1.1025 = 5000
Q21. B (62500)
P = 78732 / (1.08³)
= 78732 / 1.2597 = 62500

Q22. B (12%)
Growth ≈ 2,24,720 / 2,00,000 ≈ 1.1236
⇒ r ≈ 12%
Q23. A (1404)
1000×1.12³ ≈ 1404
Q24. B (14049)
10000×1.12³ ≈ 14049
Q25. B (10%)
1.1³ = 1.331
অতএব r = 10%
Q26. C (≈14%)
(1+r)⁵ = 2
Trial দিলে r ≈ 14%
Q27. C (1105)
5000×1.105² − 5000
= 6105 − 5000 = 1105
Q28. A (1000)
P = 1254 / (1.08³)
= 1000
Q29. A (56000)
P = 61740 / (1.05²)
= 56000
Q30. A (10000)
P = 12100 / (1.1²)
= 10000